MUSTAFA ABUELQUMSAN

Senior Researcher Scientist

- Stony Brook University Hospital (SBU)
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https://github.com/elqumsan/

MA

EXPERIENCE

Senior Researcher Scientist
Health Equity and Medicine Resilience
Climate (HEMRC)

Stony Brook University Hospital

2025 - Present Stony Brook, USA

Senior Postdoctoral Researcher
Pain and Analgesia Research Center (SPARC) -
TANG Lab

Stony Brook University Hospital

2023 - 2025 Stony Brook, USA

Assistant Professor
Applied medical science faculty

Al-Azhar University

2020 - Present Palestine

Assistant Teacher
Life science TAGC UMR U1090
Theories and Approaches of Genomic Complexity

Aix-Marseille University

2017-2018 Marseille, France

Lecturer
Applied Medical Science faculty

Al-Azhar UNiversity

2006 - 2015 Palestine

EDUCATION

Senior Post-Doc Researcher
Pain and Analgesia Research Center (SPARC)
TANG Lab

Stony Brook University

2023 - 2025 Stony Brook, USA

Ph.D. Bioinformatics and Machine Learning
TAGC UMR U1090
Theories and Approaches of Genomic Complexity

Aix-Marseille University

2015 - 2020 Marseille, France

Master of Biosatistics
Faculty of Life Sciences

Al-Azhar University

2011- 2014 Palestine

Bachelor of Computer and Data Science
Computer engineering and information technology Faculty

Al-Azhar University

1999- 2004 Palestine

TEACHING & MENTORING EXPERIENCE

1. Teaching

Foundations of Statistics,
Statistics in Bioinformatics - Course director
Al Azhar University

2019 Palestine

Analysis Statistics,
Genomic Data - Course director
Aix-Marseille University

2017 Marseille, France

Probability & Statistics, Mathematics, and Statistics in Bioinformatics - Course director
Al Azhar University

2014 Palestine

Multivariate Analysis - Course director
Al Azhar University

2013 Palestine

2. Mentored students

Ali Abu Asi
Master's in Data Science
Al Azhar University

2021 Palestine

Reem Abu Ali
Master's in Data Science
Al Azhar University

2021 Palestine

- **RESEARCH ACTIVITY**

My research interests are multidisciplinary with emphasis on studying the Next-Generation Sequencing (NGS), Multi-omics, Metadata, and cancers diseases using analytical and computational approaches. I was partner with studies in recent years such as in relation to cancer types. My current main research interests include studying the spread of machine learning algorithms to analysis high-throughput data, multi-omics, metadata.

- Single cell RNA-seq technologies, is being a powerful tool to study the presence and quantity of RNA molecules in biological samples and have revolutionized transcriptomic studies on bulk tissues, but, recently, the emerging single-cell RNA-seq (scRNA-seq) technologies enable the investigation of transcriptomic landscapes at a single-cell resolution, wherein sequencing the DNA of individual cells can give information about mutations carried by small population of cells, that providing a higher resolution of cellular differences and a better understanding of the function of an individual cell in the context of its microenvironment (Li, 2019).
- During my PhD thesis (Abuelqumsan, 2018) I developed the RNAseqMVA R-package, which performs comparative assessment of supervised classification methods (currently supported methods: SVM, RF, KNN) to classify biological samples based on scRNAseq transcriptome profiles.
- The package compiles with the recommended best practices for reproducible science: the source code is available on github: (https://github.com/elqumsan/RNAseqMVA) A specific conda environment has been defined, which enables to deploy all the software environment (R, libraries) required to compile and run the scRNAseqMVA package (portability of the code). The package also contains yaml-formatted configuration files enabling to reproduce all the results of the study cases presented in my PhD thesis
- (reproducibility), as well as to modify the parameters in order to analyse other data sets (re-usability). We are in the process of writing a Vignette which will provide detailed information about the use of the package, and enable its usage for analyses of custom datasets. Beside that I developed R package to analysis the robustness of right and middle censoring schemes in parametric survival models (Abuzaid et al., 2017) that has been pivoted about the what is the robustness of right and middle censoring schemas in parametric survival models by it we figured out the middle censoring schema which has achieved the best performance from among the rest censoring schemas that are right and left.
- The main goal of applying is to evaluate the performances of Artificial Intelligence algorithms and then develop AI algorithms to classify cancer samples based on transcriptome, characterized by either bulk or single-cell RNA-seq sequencing. Moreover, finding the optimal pre-processing approaches that is essential for dealing with this kind of classification mechanisms. Afterward keep forward to with Multi-omics, metadata, …
- Moreover, I recently launched R package which named (RNAseqMVA) in GitHub repository (https://github.com/elqumsan/RNAseqMVA) that is concerning in Evaluation and develop of Machine Learning methods for supervised classification of NGS data – RNAseq data.

- You can follow my research processes in the LinkedIn website (https://www.linkedin.com/in/mustafa-abuelqumsan-6a585b58/)

Main applications:

In the field of snRNA-seq special transcriptome and microarray expression data, I work on the preprocessing and analysis of static and dynamic expression data. I have also expertise in the preprocessing multivariate analysis of genetic variations.

In the field of Metagenomics, I works on optimal primer design, pre-processing and analysis of sequencing data and reverse engineering of cancer networks.

In the field of count data I develop methods for the preprocessing and classification snRNAseq and NGS data.

BIBLIOGRAPHY

I. Articles in professional peer-reviewed journals

1. Abuzaid, A. H., AbuElQumsan, M. K. and El-Habil, A. M. (2015). On The Robustness of Right and Middle Censoring Schemes in Parametric Survival Models. Communications in Statistics - Simulation and Computation. 03/2015; DOI:10.1080/03610918.2015.1011337 · (ISI- Cited Publication).
2. R-package for evaluation of supervised classification methods to analysis RNAseq data. (https://github.com/elqumsan/RNAseqMVA)
3. AbuElqumsan, M., Ghattas, B., Van Helden, J. (2019). comparative assessment of the efficiency of machine-learning algorithms for the classification of RNAseq data. *In prep.*

II. Conferences/ Seminars/ Workshop

1- AbuEl-Qumsan, M. K. "Detection of influential observations in Parametric models", Seminar on Directional Data, 10 March 2010. Faculty of Mathematical Sciences, Al Azhar University.

2- AbuEl-Qumsan, M. K. On The Robustness Of Censoring Schemes With Parametric Survival data, 2 April 2014, Faculty of Mathematical Sciences, University of Al Azhar University.

3- Abuzaid, A. H. and AbuEl-Qumsan, M. K. " study the determination of the parameters of the middle censoring distributions for managing the proportion censoring". The 5th Mathematics and Physical science graduate congress, Al Azhar University, West Bank, Palestine, 7-9 December 2009.

4- Abuzaid A. H. and AbuEl-Qumsan, M. K. "develop comprehensive R- package for the analysis of middle censored data.". 17th National Symposium of Mathematical Sciences. West Bank, Palestine, 15-17 December 2009.

HONOR AND AWARDS

Name of award	***Date awarded***
Islamic Development Bank (IDB)- Jeddah for Doctoral study in France	2015-2019
Al-Fakhoora International Masters Scholarships	2011-2014

SKILLS

Programming skills

Advanced R, Advanced Linux
Advanced Python, Advanced SAS
Advanced SPlus

Designing networks infrastructure
Cisco Certified Network Association (CCNA)

REFERENCES

Prof. Judi Brown Clarke	Vice President for Equity & Inclusion Chief Health Equity Officer Office of the President Stony Brook University Office: 631.632.6975 Cell: 631.332.6502 judith.b.clarke@stonybrook.edu Judith.Clarke1@stonybrookmedicine.edu

Prof. Dr. Jacques Van Helden	Co-director for Institut Français de Bioinformatique (IFB) and Full Professor in Bioinformatics Aix-Marseille Université (AMU). Lab. Theory and Approaches of Genomic Complexity (TAGC) INSERM Unit UMR_S 1090, 163, Avenue de Luminy, 13288 MARSEILLE cedex 09. France Office: INSERM building, block 6 Tel: +33 4 91 82 87 49 Fax: +33 4 91 82 87 01 Web: http://jacques.van-helden.perso.luminy.univ-amu.fr/ http://www.france-bioinformatique.fr https://elixir-europe.org/about-us/who-we-are/nodes/france Email: Jacques.van-Helden@univ-amu.fr
Prof. Badih Ghattas	Full Professor in Mathematics and statistics Department Aix-Marseille University Marseille, France Email: badih.ghattas@univ-amu.fr +33 6 14 10 01 20
Prof. Ismail Zahed	Professor at physics department - Center for Nuclear Theory Stony Brook University Email: ismail.zahed@stonybrook.edu 631-632 8129 C-141

LANGUAGES

English
Advanced

French
Advanced

Arabic
Advanced